

Libraries: NumPy & Pandas

Q1. Load the dataset and display the first 5 rows with dataset information

```
import pandas as pd
import numpy as np

# Load dataset
df = pd.read_csv("retail_dataset.csv")

# Display first 5 rows
print(df.head())

# Dataset information
print(df.info())
```

Output Summary

- Rows: **100**
 - Columns: **7**
 - Numerical Columns: Quantity, Price
 - Object Columns: Transaction ID, Customer ID, Product Category, Date, Region
-

Q2. Identify missing values in each column

```
print(df.isnull().sum())

print("Total Missing Values =", df.isnull().sum().sum())
```

Output

Column	Missing Values
Transaction ID	0
Customer ID	0
Product Category	3
Quantity	2
Price	5
Date	0

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Region 2

Total Missing Values = 12

Q3. Replace missing values in Quantity and Price with mean

```
df["Quantity"] = df["Quantity"].fillna(df["Quantity"].mean())
df["Price"] = df["Price"].fillna(df["Price"].mean())
```

Q4. Remove rows where Product Category or Region is missing

```
df = df.dropna(subset=["Product Category", "Region"])
```

Q5. Create Revenue column

Formula:

Revenue = Quantity × Price

```
df["Revenue"] = np.multiply(df["Quantity"], df["Price"])

print(df.head())
```

Q6. Calculate total revenue

```
total_revenue = np.sum(df["Revenue"])

print(total_revenue)
```

Output

Total Revenue = 465529.66

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Q7. Revenue by Product Category

```
category_revenue = df.groupby("Product Category")["Revenue"].sum()

print(category_revenue)
```

Output

Product Category	Revenue
Beauty	27647.00
Books	29597.94
Clothing	85006.00
Electronics	93204.32
Furniture	129664.53
Groceries	11978.00
Sports	34484.00
Toys	53947.88

Q8. Top 3 and Bottom 3 Product Categories

```
top3 = category_revenue.sort_values(ascending=False).head(3)

bottom3 = category_revenue.sort_values().head(3)

print(top3)
print(bottom3)
```

Top 3 Categories

Category	Revenue
Furniture	129664.53
Electronics	93204.32
Clothing	85006.00

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Bottom 3 Categories

Category	Revenue
Groceries	11978.00
Beauty	27647.00
Books	29597.94

Q9. Revenue by Region

```
region_revenue = df.groupby("Region")["Revenue"].sum()
```

```
print(region_revenue)
```

```
print("Highest Revenue Region")  
print(region_revenue.idxmax())
```

```
print("Lowest Revenue Region")  
print(region_revenue.idxmin())
```

Output

Region	Revenue
North	153244.35
East	126232.04
South	104855.74
West	81197.53

Highest Revenue Region: North

Lowest Revenue Region: West

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Q10. Monthly Revenue + Mean, Median & Standard Deviation

```
df["Date"] = pd.to_datetime(df["Date"], dayfirst=True)

df["Month"] = df["Date"].dt.month_name()

monthly_revenue = df.groupby("Month")["Revenue"].sum()

print(monthly_revenue)

print("Mean =", np.mean(df["Revenue"]))
print("Median =", np.median(df["Revenue"]))
print("Standard Deviation =", np.std(df["Revenue"]))
```

Monthly Revenue

Month	Revenue
January	43994.00
February	33028.00
March	10557.00
April	57298.53
May	27915.00
June	28190.00
July	47851.29
August	76958.00
September	30719.11
October	38017.00

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November 27662.7
4

December 43339.0
0

Revenue Statistics

- **Mean:** 4900.31
- **Median:** 3282.11
- **Standard Deviation:** 4721.17